

BACH PHAM

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Education

True North International School	Hanoi
	Sep 2024 -
	June 2027
STANDARDIZED TESTS	
IELTS:	xxx 202x
SAT:	xxx 202x
AP Microeconomics:	5/5
AP Macroeconomics:	5/5
AP Precalculus:	5/5
AP Computer Science A:	4/5

Honors & Awards

Gold Medal - INOVA 2025, 49th International Invention Show	Croatia
- Project: Exchange Rate Risk for Vietnamese Import-Export Enterprises. - Developed a quantitative framework to analyze currency fluctuations and proposed financial hedging strategies for trade enterprises.	2025
Gold Medal - International Creative Papers Conference (ICPC)	Korea
- Project: Detection via Deep Neural Variational Autoencoder Oblique Random Forest. - Engineered a hybrid machine learning model to automate anomaly detection and enhance risk management in high-dimensional datasets.	2025
Gold Medal World Invention Creativity Olympics (WICO)	Korea
Analyzed FDI influx drivers and policy frameworks in Hanoi's tech sector, formulating strategic policy recommendations for Hoa Lac Hi-Tech Park.	Jul 2026
Gold Award AX Global Olympiad	Korea
Conducted a comprehensive business analysis to design a data-driven, multi-channel marketing strategy for GTE Localize's LocReset platform.	Jul 2026

Research Experience

Title: Performance Optimization in Healthcare Analytics via Explainable AI (XAI) - [Link]	May 2025
Published in the IJMSM, Vol. 2, Issue 4, 2025	

Key Contributions:

- Data Engineering & Imbalance Handling:** Integrated and preprocessed diverse datasets using SMOTE and ADASYN techniques to mitigate class imbalance, ensuring high-quality data inputs for robust financial and healthcare modeling.
- Predictive Analytics & Model Fusion:** Evaluated multiple machine learning classifiers, including XGBoost and Random Forest, using GridSearchCV for hyperparameter tuning to achieve a peak prediction accuracy of 83%.
- Interpretable Business Insights:** Leveraged SHAP and LIME frameworks to transform "black-box" model outputs into transparent insights, identifying key risk drivers such as Glucose and BMI to support data-driven strategic decisions.

Personal Project

Project name: Automated Crypto Trading Bot with Technical Analysis	May 2025 - Jan 2026
Key Features / Highlights:	
Algorithmic Trading & Strategy Development: Engineered an automated trading system utilizing Python and the CCXT library to execute real-time trades on global cryptocurrency exchanges. Quantitative Technical Analysis: Integrated multiple technical indicators, including Moving Averages (MA), Relative Strength Index (RSI), and MACD, to identify optimal market entry and exit points. Risk Management & System Architecture: Implemented a robust backtesting framework and modular architecture using environment variables (.env) to ensure API security and strategy optimization.	

Social Work & Activities

Performing Arts – Violinist	2020
Second Prize Muse 2020 International Music Competition: Awarded second place in classical violin performance within a global competitive field. Special Prize Malta International Music Competition: Recognized by an international jury for exceptional musicality and technical performance.	
Community & Volunteer Work	2025
Volunteer "Warm Winter" (Mùa Đông Ấm) 2025 Campaign: Coordinated resource collection and distributed winter supplies to support underprivileged families in rural areas.	
School Athletics – Football & Badminton Clubs	2025
Active Member: Represented the school in inter-school friendly matches and regular training sessions, enhancing physical fitness and team collaboration.	

Core Skills

Financial Analysis: Algorithmic trading (Python, CCXT), technical indicators (RSI, MACD), and quantitative risk assessment.
Data Science: Predictive modeling (XGBoost, Random Forest), Explainable AI (SHAP, LIME), and data preprocessing (SMOTE, ADASYN).
Technical Tools: Proficient in Python (Pandas, Scikit-learn), SQL, Flask backend, and Firebase real-time databases.